

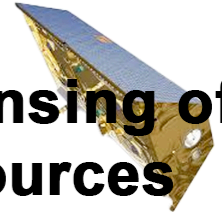


Australian
National
University

Research School
of Earth Sciences



**Remote Sensing of
Water Resources**
EMSC3025/6025



Water Cycle

EMSC3025/6025

Objectives of this Lecture

By the end of this lecture, you should be able to:

- Describe the **structure and properties of water** relevant to hydrology
- Understand the **concept and components of the hydrological cycle**
- Explain the **global distribution of water** and its limited availability
- Define the **water balance equation** and interpret each term
- Be able to recognise a **surface catchment**
- Identify challenges in applying hydrological concepts at different scales

What is hydrology?

- *Hydrology* comes from Greek *hydor* (water) and Latin *logia* (study).
- Despite the name, it focuses on **fresh water**, not all water.
- **Oceanography** is the science of saline water.
- Hydrology examines:
 - Distribution and movement of water
 - Interactions with land, atmosphere, and ecosystems

The oldest known description of water cycle:

[...] precipitation falling in the mountains infiltrated the Earth's surface and led to streams and springs in the lowlands. [...]

— *Vitruvius, Roman Architect, 1st century BC*

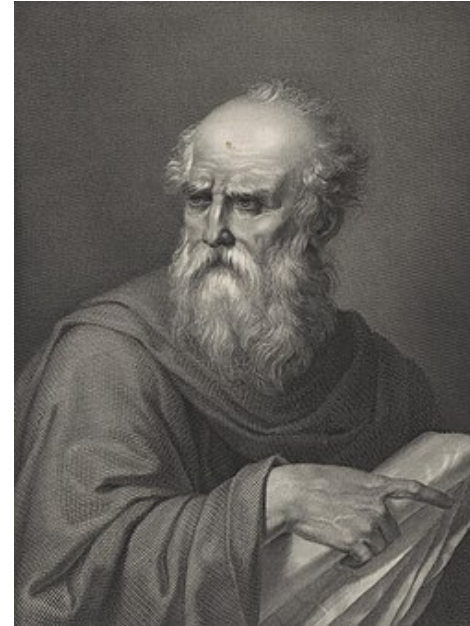


Figure 1: Hydrology is one of the oldest sciences. Egyptians, Mesopotamians, Greeks and Chinese needed to understand water cycle to build irrigation systems. The oldest known description of water cycle is *Vitruvius*

Humans and water



Primary dimension of human interference

Land use

Climate change

Water use

Hydrology as a science

“The total quantity of fresh water on earth could satisfy all the needs of the human population if it were evenly distributed and accessible.” — Stumm (1986)

- Hydrology explains **uneven water distribution**
- Serves as:
 - A **pure science** – understanding natural processes
 - An **applied science** – solving water challenges



Figure 2: Resolving global disparities in water availability and demand is one of the key SDGs

UN's 17 Sustainable Development Goals

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Disciplinary pathways

ETH-Zurich

- Two main traditions:
 - **Engineering hydrology** – design-focused, quantitative
 - **Earth science hydrology** – process-focused, explanatory
- This course adopts a **quantitative Earth science** perspective
- Related fields:
 - **Geohydrology** – groundwater systems
 - **Ecohydrology** – water–ecosystem interactions



Figure 3: We will focus on water processes in this course. Hydraulic engineering is more focused on design and measurement focus.

A resource and more

- Covers more than **70%** of Earth's surface
- Essential to:
 - **Human survival**
 - **Agricultural production**
 - **Ecological functioning**
- For **Indigenous Australians**, water is integral to **Country**, ceremony, and responsibility

Up the river

by Margaret Simons

Politics



Hope is running dry in the Murray–Darling Basin

Figure 4: An article in *the Monthly* magazine on the fate of the Murray-Darling Basin

Watch **Homily to Country** — National Gallery of Victoria

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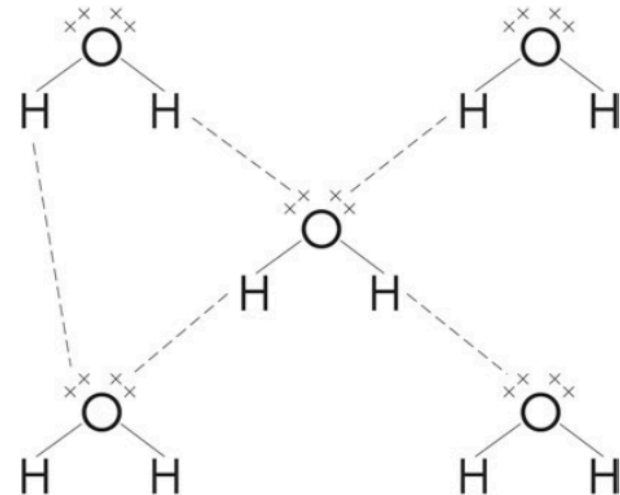
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Physical properties of water

Molecular properties of water

- Water: H_2O two hydrogen atoms covalently bonded to oxygen
- It is a **bipolar molecule**:
 - Positive hydrogen, negative oxygen
- Bonds:
 - **Covalent** within molecules
 - **Hydrogen** between molecules
- Leads to:
 - High **surface tension**
 - Strong **cohesion**
 - Exceptional **solvent** capability



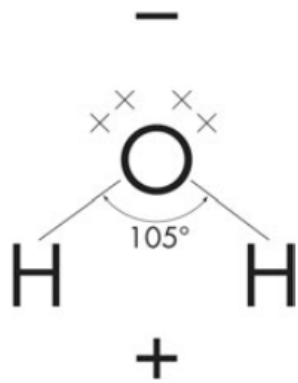


Figure 5: Structure of a water molecule

Figure 6: Hydrogen bonding between water molecules

Physical Properties

Density

- Water is most dense at 4°C
- Ice is **less dense** → floats on water
- Implications: - Lakes freeze from top-down - Aquatic ecosystems persist in cold conditions

Specific Heat

- Water has **high specific heat capacity** (4.2 kJ/kg/K)
- Slows temperature change
- Buffers **climate** and daily temperature swings
- Comparison with other materials:

Substance	Specific Heat (kJ/kg/K)
Water	4.2
Dry soil	1.1
Ethanol	0.7
Iron	0.44

Table 1: Specific heat capacity of different substances

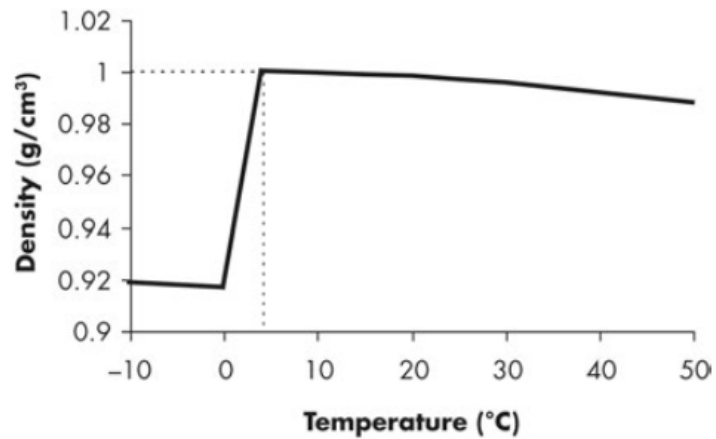


Figure 7: Water density as a function of temperature

Phase transitions

- The high energy required to break hydrogen bonds makes water an ideal climate stabiliser.
- Water changes state:
 - Solid ↔ Liquid ↔ Gas
- Processes include:
 - Melting, freezing
 - Evaporation, condensation
 - Sublimation, deposition
- Involves latent heat
- Crucial for energy transfer in climate systems

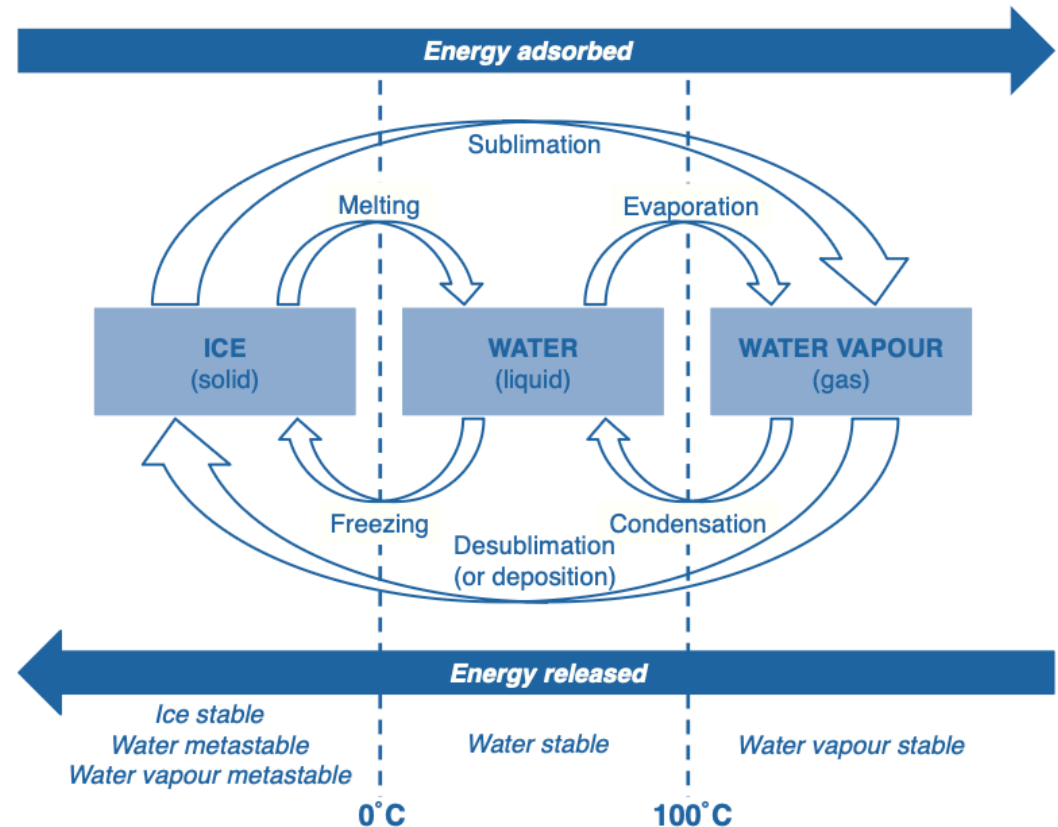


Figure 8: Phase transitions and latent heat of water

Let's switch from micro to macro! Where do we look?

A spatial unit to study the water cycle?

A catchment?

- A **catchment** is a land area where surface water drains to a common point
- Defined by **topographic divides**
- Used to define **hydrological boundaries**
- Contains **nested sub-catchments**

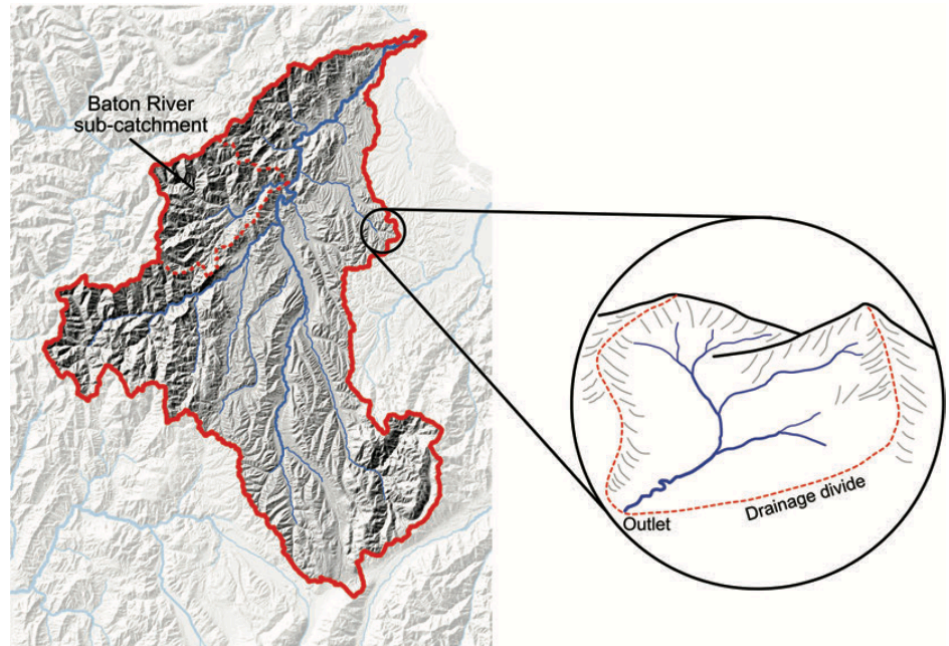


Figure 9: Catchment boundaries and surface drainage patterns

Groundwater vs surface catchments

- **Surface water divides $\neq$ groundwater divides**
- Groundwater can flow across topographic boundaries
- Important for:
 - **Integrated water resource management**
 - Understanding flow connectivity

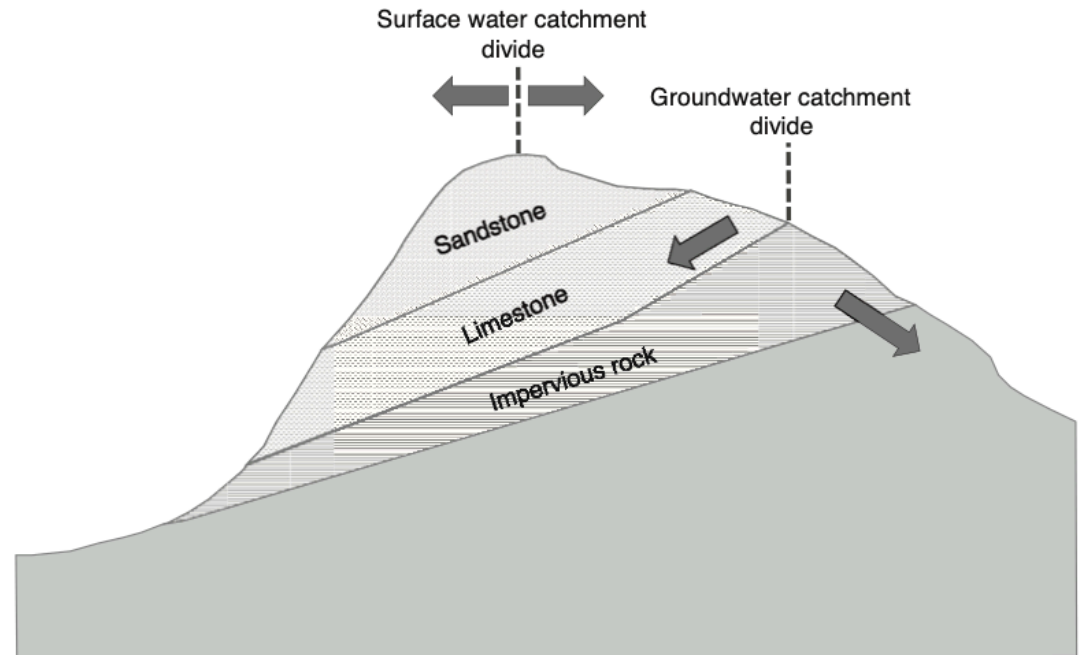


Figure 10: Difference between groundwater and surface water divides

How to identify a catchment?

1. Get a topographic map

- A contour map with visible elevation lines.
- Mark known streams (especially the outlet point).

2. Identify the outlet (pour point)

- The point you want to delineate the catchment (where stream exits).

3. Mark all flow lines (ridges to valleys) - Draw approximate flow paths down the steepest slope.

4. Trace ridgelines (water divides)

- Trace along high points (ridges) that separate your catchment from neighbouring ones.
- Follow the highest elevation between two drainage areas.
- Cross contour lines at right angles (steepest ascent).
- Close the boundary loop at the other side of the outlet.

5. Check connectivity

- Ensure all water within your boundary flows to the outlet.

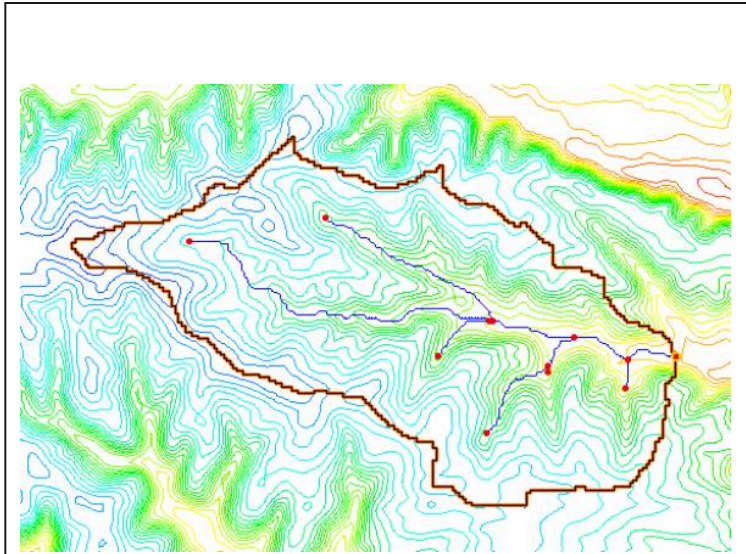


Figure 11: Catchment identification

The hydrological cycle

- The **hydrological cycle** describes water movement between Earth and atmosphere as gas, liquid, or solid.
- It is a conceptual model — useful but simplified.
- Hydrology begins with the **global scale**, then zooms into **catchments**.

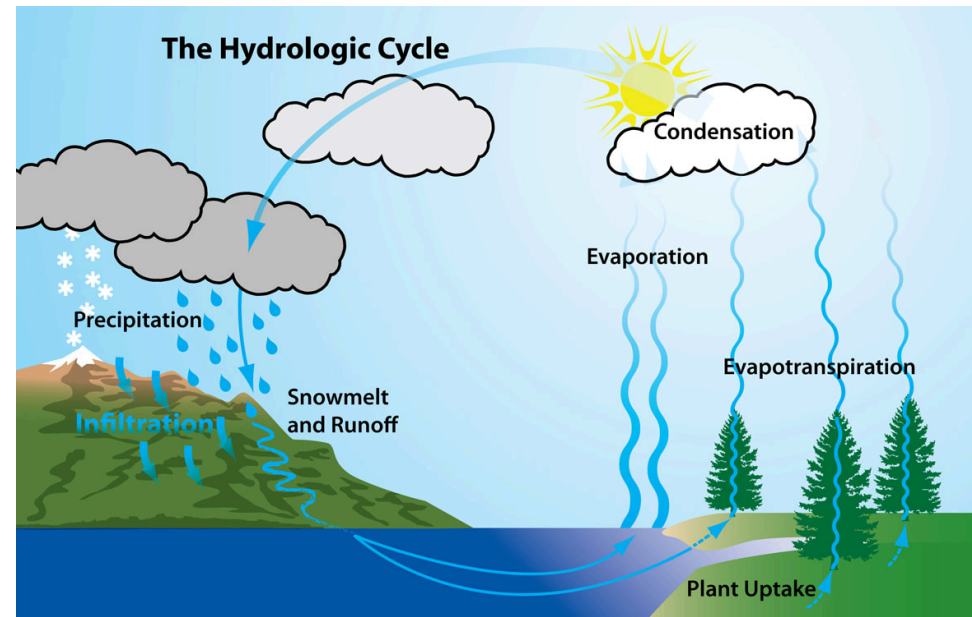


Figure 12: Different elements of the hydrological cycle

Global distribution of water

- Most of Earth's water is stored in:
 - Oceans/seas: **96.5%**
 - Ice/glaciers: **1.74%**
 - Groundwater: **1.69%**
- Rivers and lakes make up **tiny fractions** of total volume.

groundwater with the 1st km of the surface, and discount ice and snow $\Rightarrow$ we have **0.27%** of which is accessible.

Q: Is this enough water per person?

Storage	Volume ($\times 10^3 \text{ km}^3$)	% of total
Oceans/seas	1,338,000	96.54
Ice caps/glaciers	24,064	1.74
Groundwater	23,400	1.69
Permafrost	300	0.022
Lakes	176	0.013
Soil	16.5	0.001
Atmosphere	12.9	0.0009
Marsh/wetlands	11.5	0.0008
Rivers	2.12	0.00015
Biota	1.12	0.00008

Table 2: Global estimates of water distribution

Components of the global cycle

- Key global processes:
 - **Evaporation** from oceans/lakes
 - **Precipitation** over land and sea
 - **Run-off** moves water back to oceans
- Oceans evaporate more than they receive.
- Continents receive more precipitation than they lose.
- Ironically the terrestrial component of the cycle is by far the smallest.

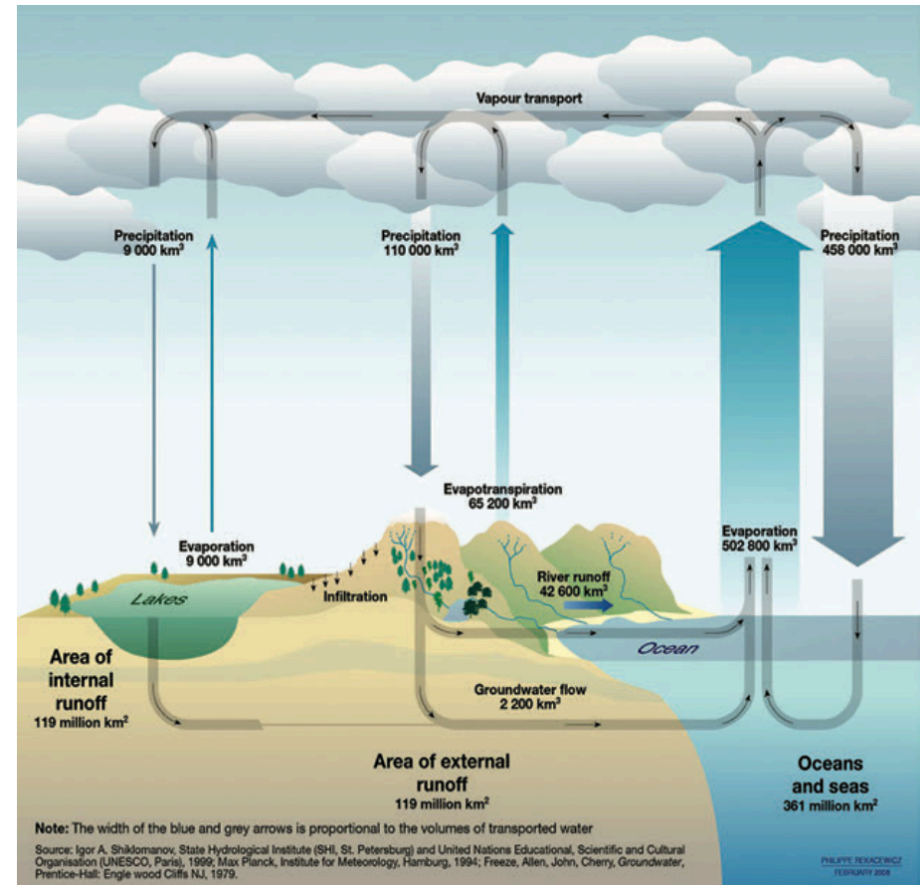


Figure 13: The global hydrological cycle and proportional flow volumes, Rekacewicz, UNEP 2008

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Climate zones and water partitioning

- Precipitation is partitioned into:
 - Evaporation
 - Run-off
 - Groundwater recharge
- Distribution varies by climate:
 - Humid temperate: balanced
 - Arid zones: dominated by evaporation

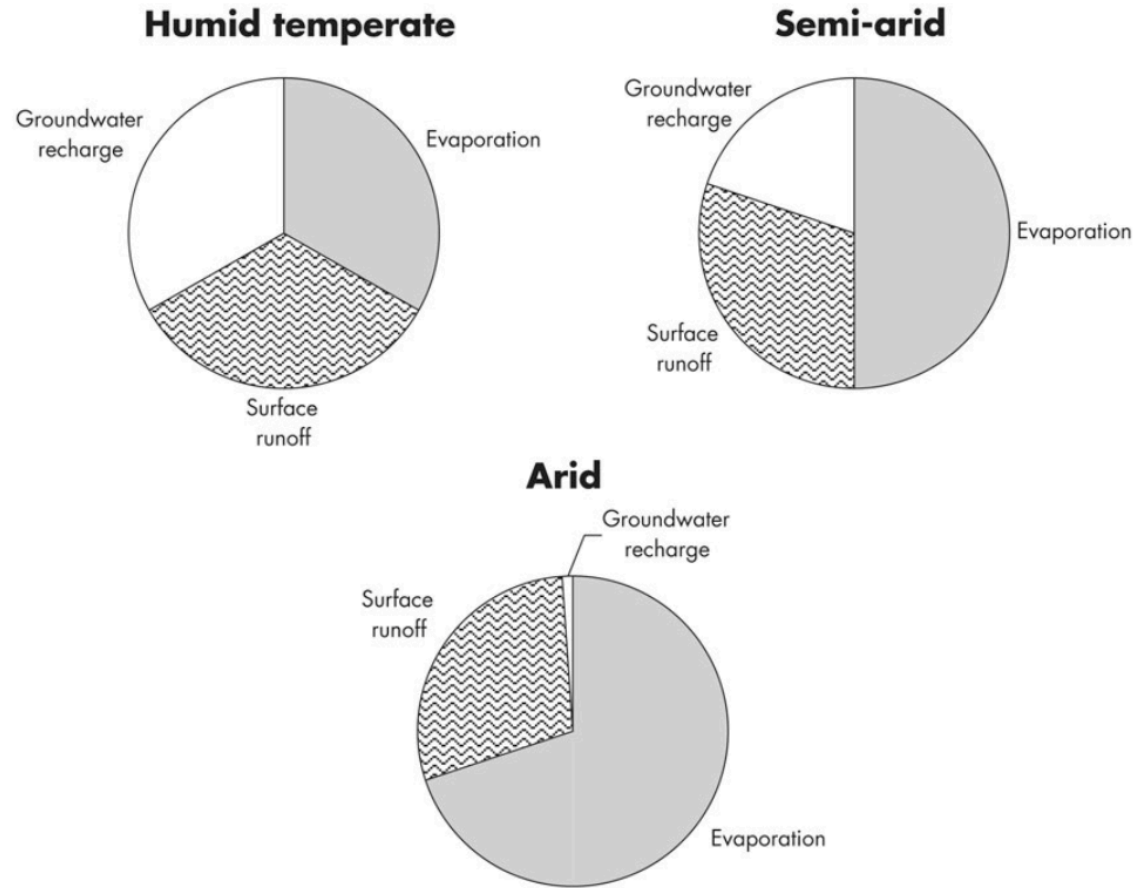


Figure 14: Partitioning of total precipitation by climate zone

Water availability vs population

146 million litres per person per year is misleading!!!

- Per capita water availability is **misleading** if population and use patterns are ignored.
- Example: Australia is water-rich by volume but highly variable in space and time.
- Effective management depends on **abstraction, storage, and equity**.

Richest countries	1,000 m ³ /yr	Poorest countries	1,000 m ³ /yr
Iceland	525	Kuwait	0
Guyana	301	Bahrain	0.003
Suriname	184	UAE	0.016
PNG	109	Egypt	0.022
Bhutan	103	Qatar	0.026
Gabon	98	Bahamas	0.053
Canada	81	Sudan	0.081
<i>Other Countries</i>			
Australia	21	South Africa	0.843
USA	8	Kenya	0.467
USA	2	Israel	0.093

Table 3: Comparison of annual internal renewable water per capita.

Water abstraction

- OECD countries vary widely in **water abstraction**.
- USA: $\sim 1,730 \text{ m}^3/\text{person}/\text{year}$
- Often driven by:
 - **Agriculture**
 - **Industry**
 - **Irrigation infrastructure**
- Water use $\neq$ water availability

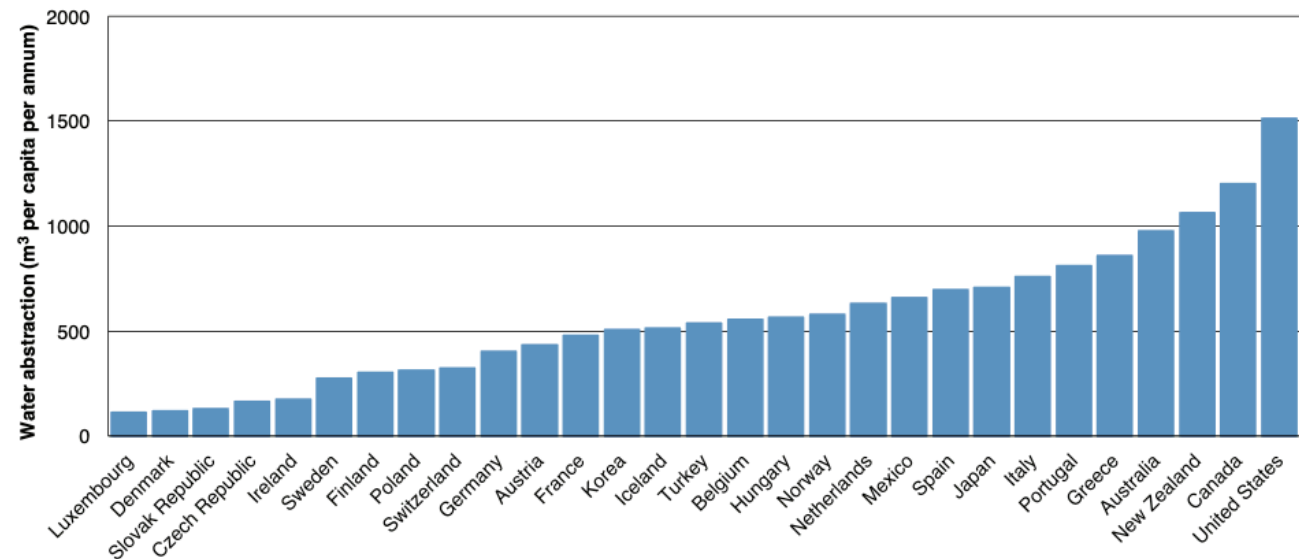


Figure 15: Water abstraction per capita for OECD countries

Groundwater Embedded in Global Food Trade

approximately eleven per cent of non-renewable groundwater use for irrigation is embedded in international food trade, of which two-thirds are exported by Pakistan, the USA and India alone.

— Dalin et al (2017), *Nature*

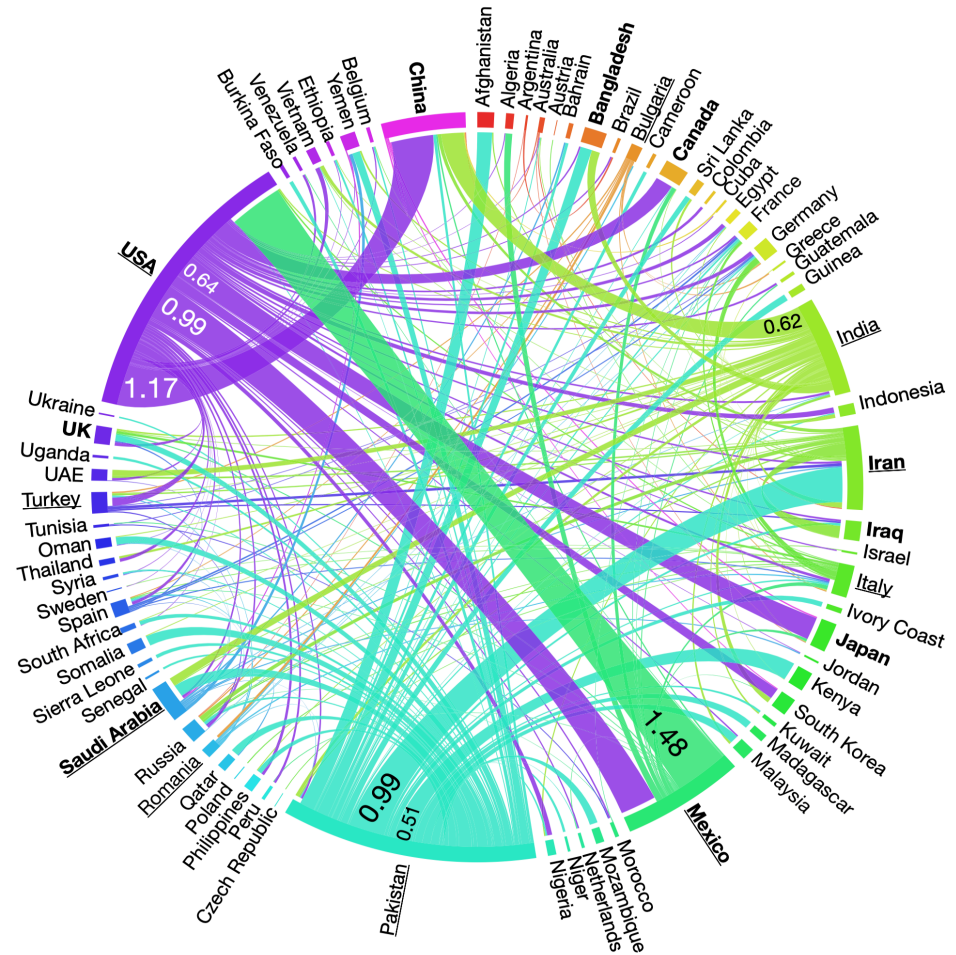


Figure 16: Groundwater Exchange Embedded in Global Food Trade

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The consequence

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The consequence

The catchment hydrological cycle

- Focuses on processes at **basin scale**:
 - Evaporation
 - Precipitation
 - Run-off
- Includes sub-processes:
 - Interception
 - Transpiration
 - Infiltration and through-flow

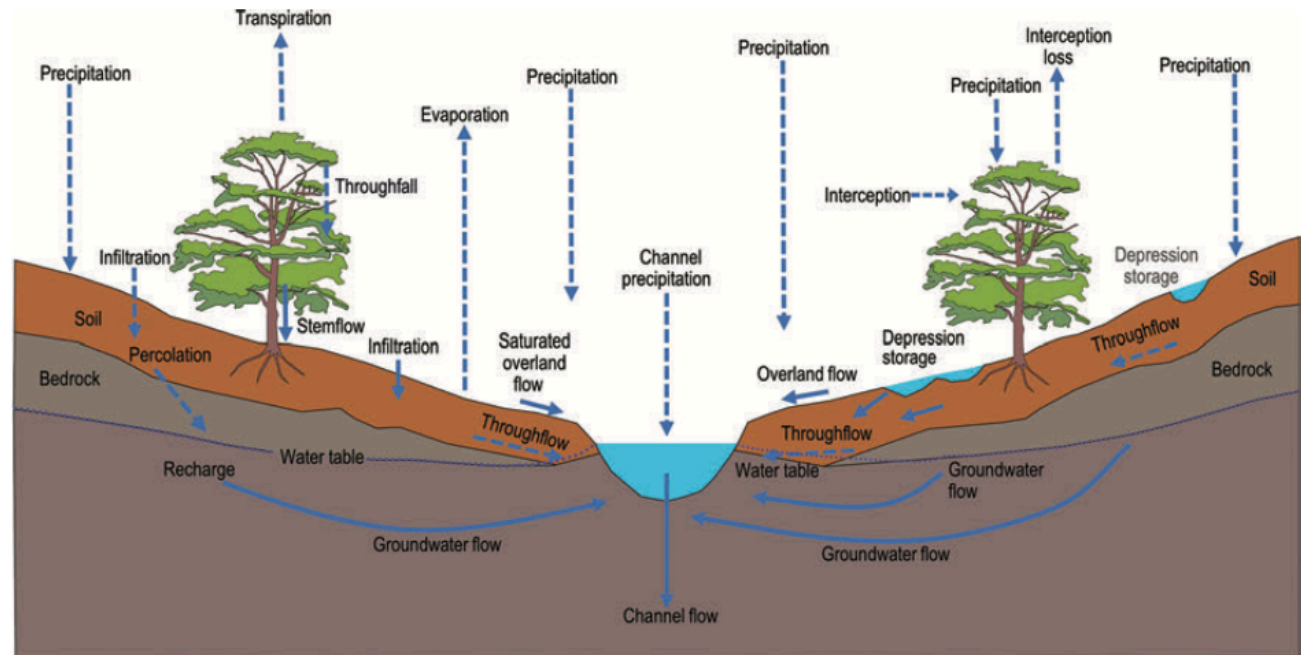


Figure 17: Hydrological cycle processes at the catchment scale

The water balance equation

- The **water balance equation** represents the continuity of water mass in a system.
- It quantifies water input, output, and storage over time.

$$P \pm E \pm \Delta S \pm Q = 0$$

- Often rearranged to estimate **streamflow (Q)**

$$Q = P - E - \Delta S$$

Where:

- P : Precipitation
- E : Evaporation
- ΔS : Change in storage

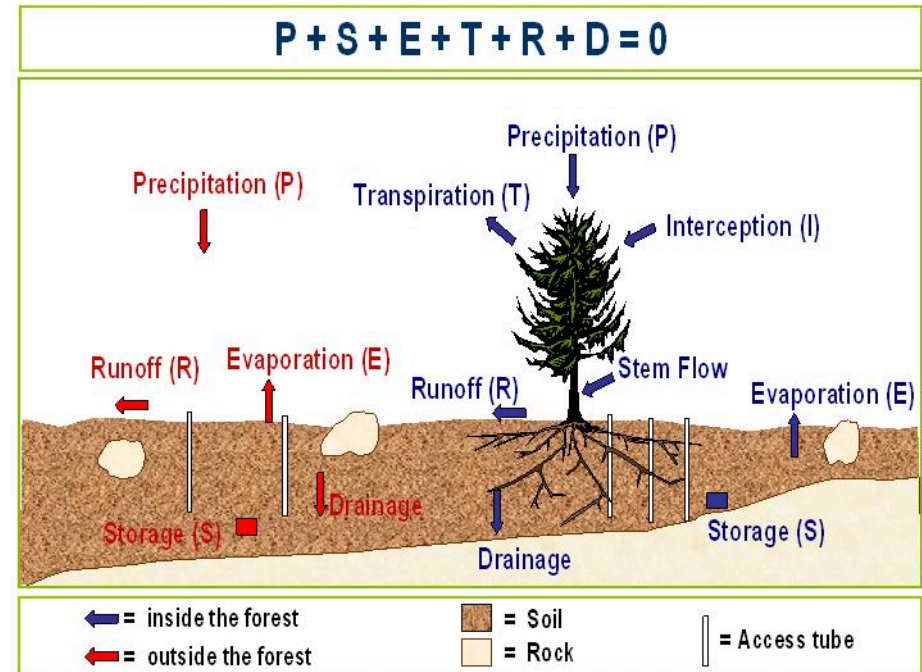


Figure 18: Mass conservation is the primary concept in the water balance equation.

Understanding the terms

- The equation includes both **fluxes** and **stores**:
 - **Fluxes:** P, E, Q
 - **Store:** ΔS (soil, groundwater, snow)
- Each term may be:
 - **Positive** (gain) — e.g. precipitation
 - **Negative** (loss) — e.g. evaporation or outflow
- Storage can increase or decrease depending on the balance.
- Used widely in:
 - **Catchment hydrology**
 - **Water resource models**
 - **Climate impact studies**
- Knowing three of the four terms allows you to estimate the fourth.
- Example: If $P = 100$, $E = 40$, $\Delta S = 10$, then $Q = 100 - 40 - 10 = 50$

Challenges in practice

- Difficulties in application include:
 - **Spatial variability** in rainfall
 - **Temporal mismatch** in data resolution
 - Estimating ΔS is often hard
- Hydrological **models** often use the water balance to simulate run-off.
- Model example: Input daily rainfall and evaporation to calculate daily discharge using: $Q = P - E - \Delta S$

Magnitude–frequency–duration

- Hydrological events vary by:
 - Magnitude** (e.g. rainfall depth)
 - Frequency** (how often)
 - Duration** (how long)
- Frequency histograms help visualize event rarity
- Important for flood design and infrastructure planning
- Probability: $p = \frac{n}{N}$
- Recurrence interval = $(1/p)$
- A 1% chance event = 1-in-100-year event
- Recurrence $\neq$ prediction of exact timing
- Used for rainfall design curves and flood risk

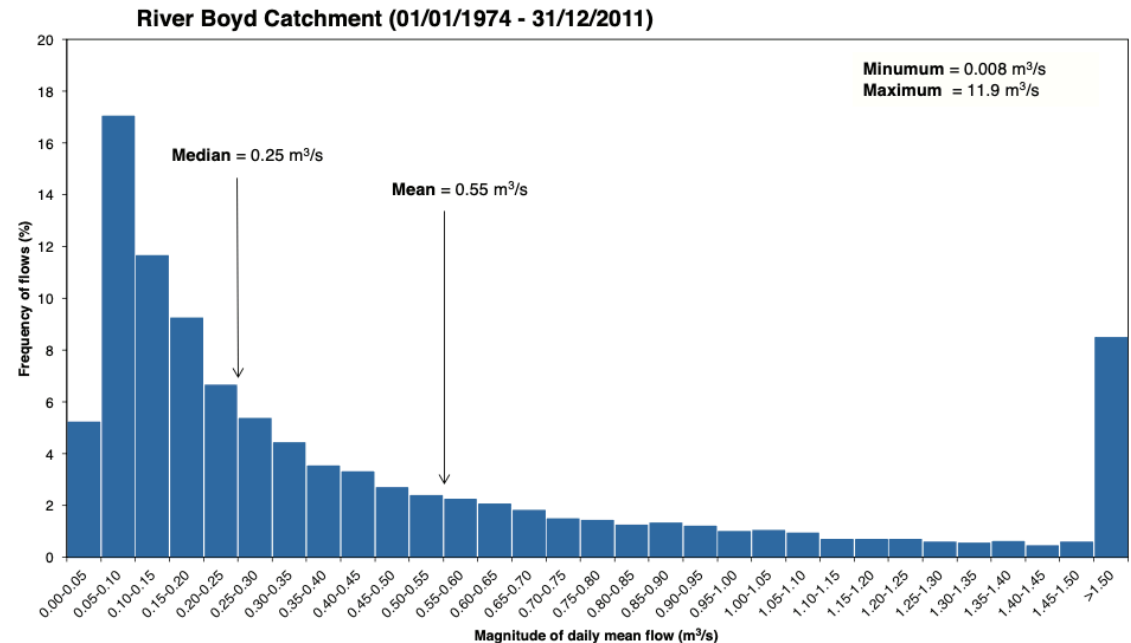


Figure 19: Flow magnitude and frequency in River Boyd catchment

Duration and design rainfall

- Magnitude–frequency–duration (MFD) curves relate:
 - Rainfall depth
 - Storm duration
 - Event rarity
- Longer durations require more rainfall to be “rare”
- Used to develop **intensity–duration–frequency (IDF)** charts

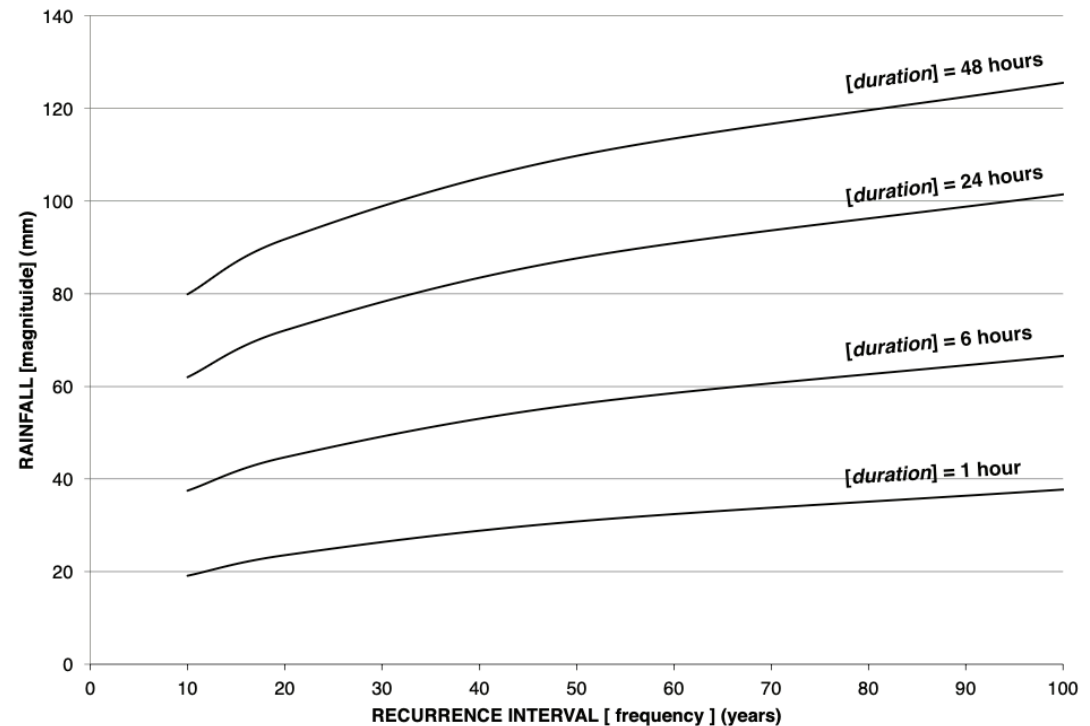


Figure 20: Rainfall MFD curves for the River Boyd catchment

Summary

In this lecture, we covered:

- The **definition and scope of hydrology** as a science of fresh water
- The **physical and molecular properties** of water that influence climate and flow
- The structure and function of the **hydrological cycle**, from global to catchment scale
- Key concepts such as **catchments, water availability, and flux vs storage**
- The **water balance equation** as a tool for understanding and modelling water movement
- Challenges in **quantifying water fluxes** and interpreting hydrological variability